



First ever spacecraft to trojans

NASA's Lucy mission will be first ever spacecraft to visit Jupiter's Trojan asteroids.
<https://digit.in/oct21-51>



YT cracks on anti-vaccine videos

YouTube cracks down on anti-vaccine videos, bans major accounts.
<https://digit.in/oct21-52>

- Zoom in to the level of detail required with your mouse scroll wheel on the wave pattern.
- Now, enable the Spectral Frequency Display by clicking the corresponding icon on the toolbar or simply press Shift + D. This opens a spectral pattern that helps in easy noise pattern identification.
- The beginning of the clip is usually a good location to identify noise. You may also want to check towards the end of the file as well. The Spectral Frequency Display shows noise as a purple pattern.

Pro Tip: Recording a few seconds of ambient noise in the beginning makes it easy to identify the exact noise pattern and apply it to the entire clip.

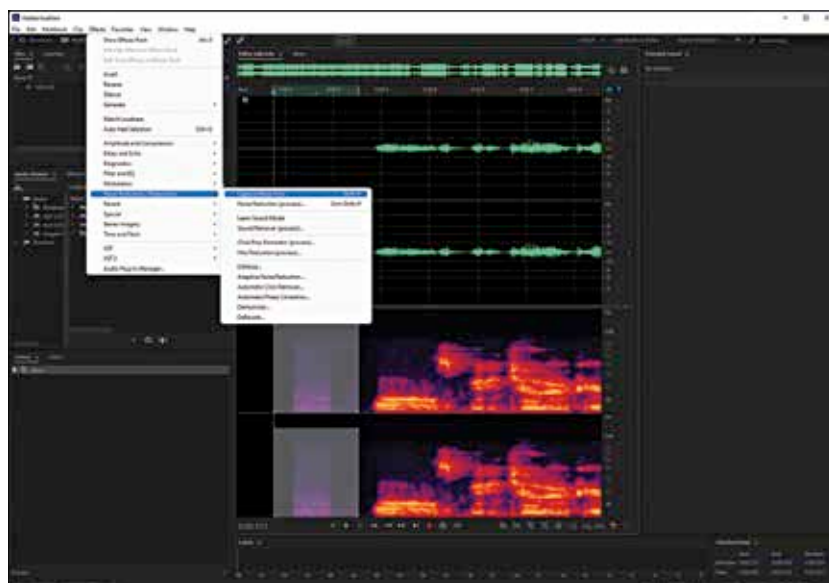
- Now that we have identified the noise pattern, it makes sense to capture this “noise print” so that it can be applied to whole track. First, drag your mouse around an area of the noise pattern. Then go to the Effect menu, navigate to the Noise Reduction / Restoration submenu, and select “Capture Noise Print” or just press Shift + P. Click Ok on the pop-up that comes up.

Pro Tip: It's recommended to select a noise pattern that is as homogeneous as possible without any additional clicks or pops in between. This way, you can ensure that you are removing just the background noise and not any audio frequency that may potentially overlap with your actual recording.

- Now, we would want to subtract this noise pattern from the whole clip. To do so, select the entire track by pressing Ctrl + A (or Command + A on Mac) and once again navigate to the Noise Reduction / Restoration submenu and click on “Noise Reduction (process)...” or press Ctrl + Shift + P to open the Noise Reduction dialog.
- The Noise Reduction dialog shows the noise floor, so you get an idea of the minimum and maximum amplitudes of the noise. There are several options to tinker with here, but for this exercise, we are primarily concerned with the Noise

Reduction and Spectral Decay Rate options. Increase the Noise Reduction slider to 100% and reduce Spectral Decay Rate to 0.

- You can also use the “Show Preview Editor” toggle to see a live preview of the effect. You should be able to see a difference in the noise pattern in the preview pane. Play back the audio to see if the noise reduction has been satisfactory. Do note that preview time may be high depending on the size of the audio track and the speed of your computer.



Capturing the noise pattern

- You can tweak the values in the Noise Reduction dialog further or even set a threshold in the graph if you feel that the program has cleaned more noise that it ought to have. It may take some experimentation to get the right values, but the effort is worth it.

DEALING WITH CLIPPING

Clipping is a type of audio distortion that occurs when the waveform hits or crosses the upper amplitude boundary. Depending on your project, clipping may be good or bad and is more of a subjective perception. But generally, it is best to avoid clipping to ensure a more pleasant listening experience.

- Navigate the waveform and see how frequent the clipping is. Play back the clipped portions and observe if those seem unpleasant. Your judgement matters a lot here.
- A more convenient way would be to go to the Window menu and clicking on “Amplitude Statistics”. This opens a dialog box with all kinds of amplitude-related parameters (if you do not see the window, ensure that it is not docked in any of the panels to the left) for all channels. Look out for the “Possibly Clipped Samples”

value to see the number of clip-pings in the track.

Pro Tip: It's always recommended to be wary of clipping in the recording stage itself. Observe the sound levels in your DAC or audio recording software and avoid those that cross the green/yellow range into the red.

- Once you decide to go ahead with reducing clipping, go to the Window menu and click on Diagnostics. This opens a Diagnostics panel on the left.
- In the panel, select “DeClipper” in the Effect dropdown and “Restore Heavily Clipped” in the Preset dropdown. Then click on Settings.
- You would want to now tune the Gain and Tolerance values to suit